# CLARA POLLOCK

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Copenhagen, Denmark


## EDUCATION

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|---|---|
| **2024 – 2028** | **PhD in astronomy** — Cosmic DAWN Center, Niels Bohr Institute, Denmark<br>Spectroscopic characterisation of the gas and chemical enrichment of the first galaxies<br>Supervisors: Kasper Elm Heintz & Sune Toft |
| **2019 – 2024** | **Integrated Masters, MPhys Astrophysics** — University of Edinburgh, UK<br>*First Class Honours*<br>Thesis: Strong-line metallicity calibrations for high-redshift galaxies<br>Supervisors: Fergus Cullen & Karla Arellano-Córdova |
| **2022 – 2023** | **Erasmus exchange** — Lund University, Sweden |

## RESEARCH INTERESTS

Early galaxy formation, spectroscopy, chemical abundances and enrichment, Lyman-$\alpha$ absorption, and reionisation.

## PUBLICATIONS

Nineteen papers, three of which are first-author. A total of 938 citations (August 2026). A full list of publications can be seen at ADS

First author papers:

- **C. L. Pollock** et. al. "Novel $z \sim 10$ auroral line measurements extend the gradual offset of the FMR deep into the first Gyr of cosmic time", A&A 708 (2026)
- **C. L. Pollock** et. al. "Characterising Ly$\alpha$ damping wings at the onset of reionisation: Evidence for highly efficient star formation driven by dense, neutral gas in UV-bright galaxies at $z > 9$", A&A 708 (2026)
- **C. L. Pollock** et. al. "Chemical signatures from the first stars embedded in metal-poor gas in galaxies at cosmic dawn", arXiv:2606.13078 (2026)

Selected publications with significant contributions:

- K. E. Heintz, **C. L. Pollock** et. al. "Dissecting the Massive Pristine, Neutral Gas Reservoir of a Remarkably Bright Galaxy at z = 14.179", A&A 987, 1 (2025)
- R. Naidu et. al. **incl. C. L. Pollock** "A Cosmic Miracle: A Remarkably Luminous Galaxy at zspec=14.44 Confirmed with JWST", OJAp 9 (2025)
- K. E. Heintz et. al. **incl. C. L. Pollock** "A massive, neutral gas reservoir permeating a galaxy proto-cluster after the reionization era", Nature Astronomy 10 (accepted, 2026)
- K. E. Heintz, **C. L. Pollock** et. al. "JWST spectroscopy of galaxies at $z > 10$: Damped Ly$\alpha$ absorbers reveal efficient star formation and hidden redshift biases", arXiv:2606.30787 (2026)

## SCHOLARSHIPS AND AWARDS

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| **2019 – 2024** | DHL UK Foundation Undergraduate Scholarship (£10,000 total) |
| **2022 & 2023** | Summer Vacation Scholarship — University of Edinburgh, School of Physics and Astronomy (£3,000 total) |
| **2019 – 2021** | Certificate of Merit — University of Edinburgh, School of Physics and Astronomy |
| **2016 – 2019** | Texas Instruments Excellence in STEM Award — Clydeview Academy |
| **2017/2018** | Dux Award — Clydeview Academy |

## TALKS

INVITED TALKS:

**January 2025** *SDU-Galaxy Seminar* — Syddansk Universitet, Odense
"Dissecting gas in galaxies from cosmic dawn to noon"

**June 2025** *Miracles of the Early Universe I* — Geneva, Switzerland
"Direct metallicities and neutral hydrogen in the early universe"

**February 2026** *From Dust till Dawn, Lorentz Meeting* — Leiden, The Netherlands
"From pristine gas to chemical enrichment"

**June 2026** *Miracles of the Early Universe II* — Geneva, Switzerland
"Insights into star-formation efficiency and chemical abundances at cosmic dawn"

CONTRIBUTED TALKS:

**June 2025** *DAWN Summit '25* — Copenhagen, Denmark
"Direct metallicities at $z > 9$ and neutral hydrogen at $z > 8$"

**April 2026** *Charting Cosmic Dawn* — Copenhagen, Denmark
"Insights into star-formation efficiency with damped Lyman-alpha absorbers"

**August 2026** *DAWN Summit '26* — Copenhagen, Denmark
"Pristine gas and chemical enrichment in the first galaxies"

## TEACHING & PROFESSIONAL RESPONSIBILITY

**2025 –** Teaching Assistant — Applied Statistics: From Data to Results

**2025 – 2026** Local Organising Committee — Charting Cosmic Dawn, with $\sim$ 140 participants

**2025 –** Organiser of research talks — Monthly DAWN days, Niels Bohr Institute